



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

SCORE-3.0

04-07-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

Test Type : SCORE-3.0

Test Pattern : JEE (Main)

TEST DATE : 04-07-2021

ANSWER KEY

PAPER-2

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A	A	B	D	D	C	B	B
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	B	A	D	C	A	D	D	C

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	B	C	D	A	C	D	C	C
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	D	C	D	B	A	C	B	A	C

MATHEMATICS

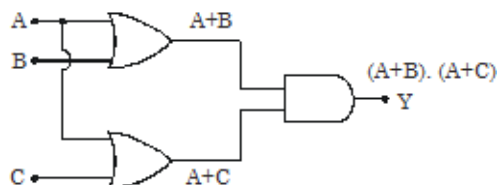
SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	C	C	A	C	A	A	A	B
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	B	B	A	B	B	C	C	A

HINT – SHEET

PHYSICS

SEC-A

1. Ans (C)



2. Ans (B)

$$\text{Radio horizon} = \sqrt{2hR_T}$$

3. Ans (A)

$$i = neAV_{\text{drift}} = \frac{V}{R}$$

$$R = \frac{\rho \ell}{A} = \frac{4\rho \ell}{\pi d^2}$$

$$\therefore V_{\text{drift}} =$$

$$\frac{V}{RneA} = \frac{V}{\frac{\rho \ell}{A} \times neA} = \frac{V}{\rho \ell ne} \propto \frac{V}{\ell}$$

4. Ans (A)

$$2k\lambda \ln \left(\frac{a}{r_1} \right) + 2k\lambda \ln \left(\frac{a}{r_2} \right) +$$

$$4k\lambda \ln \left(\frac{a}{r_3} \right) = c$$

$$= \frac{r_3^2}{r_1 r_2} = \text{const.}$$

5. **Ans (B)**

Electric field is in the direction of high potential to low potential. Potential energy = $q \times \text{potential}$.

7. **Ans (D)**

$h_b > h_d > h_c > h_a$. The liquid level is higher where the pressure is lower. The pressure is lower where the flow speed is higher. The flow speed is highest in the narrowest tube, zero in the open air.

8. **Ans (C)**

Force exerted by oil on the cylinder is zero.

$$\text{Force by water} = (4\sigma g + 4\rho g) \times A$$

$$= \left[4 \times \frac{1}{2} \times \frac{10^4}{100} + \frac{4 \times 10^3 \times 10}{100} \right] \times \frac{16\pi}{10^4} = 0.96\pi N$$

10. **Ans (B)**

$$T' = 2\pi \sqrt{\frac{\ell}{g}} = \sqrt{2} \cdot 2\pi \sqrt{\frac{\ell}{g}}$$

$$g' = \frac{g}{2}$$

$$\frac{gR^2}{(R+h)^2} = \frac{g}{2}$$

$$h^2 + 2hR - R^2 = 0$$

$$h = (\sqrt{2} - 1)R$$

12. **Ans (B)**

$$a_x = \frac{qE}{m} = 10 \text{ m/s}^2$$

$$\therefore T = \frac{2 \times 10}{10} = 2 \text{ sec}$$

$$R = 10 \times 2 - \frac{1}{2} \times 10 \times 4 = 0$$

$\therefore a_x = a_y \Rightarrow$ particle will follow a straight line.

15. **Ans (D)**

By Kepler's law

$$T^2 = \frac{4\pi^2}{GM} R^3 = \frac{4\pi^2}{G\rho} \times \frac{4\pi}{3}$$

$R = R_{\text{asteroid}}$

$\Rightarrow$ so, time period depends only on density of planet

$T = \text{TP of near earth orbit satellite} = 84 \text{ min} = 1 \text{ hr } 24 \text{ min}$

16. **Ans (C)**

We know that

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.96}{1-0.96} = 24$$

The collector current I_C is given by

$$I_C = \frac{V_C}{R} = \frac{0.5}{800} = 0.625 \times 10^{-3} \text{ A}$$

$$\text{Further } I_B = \frac{I_C}{\beta} = \frac{0.625 \times 10^{-3}}{24}$$

$$= 26 \times 10^{-6} \text{ A} = 26 \mu\text{A}$$

17. **Ans (A)**

$$R_L = 24 \times 10^0 \Omega$$

$$R_f = \frac{R_i}{2} = 12 \times 10^0 \Omega$$

18. **Ans (D)**

$$\frac{\epsilon R_1}{R_1 + R_2} = \frac{\epsilon C_2}{C_1 + C_2}$$

$$R_1 C_1 + R_1 C_2 = R_1 C_2 + R_2 C_2$$

$$\text{or } R_1 C_1 = R_2 C_2$$

19. **Ans (D)**

$$mv_0 R = mva \quad \dots(i)$$

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 - \frac{GMm}{a} \quad \dots(ii)$$

20. **Ans (C)**

Truth table for NAND gate

x	y	z
0	0	1
0	1	1
1	0	1
0	0	0

PHYSICS

SEC-B

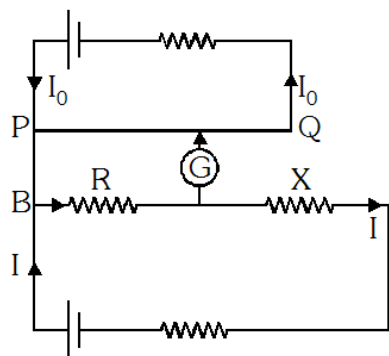
1. **Ans (4.00)**

$$\frac{1}{2} \left[\frac{q_{\text{en}}}{\epsilon_0} \right] = \phi$$

$$\phi = \frac{4q}{\epsilon_0}$$

2. **Ans (2.00)**

In potentiometer wire potential difference is directly proportional to length



Let potential drop unit length a potentiometer wire be K .

For zero deflection the current will flow independently in two circles

$$IR = K \times 10 \quad \dots(1)$$

$$IR + IX = K \times 30 \quad \dots(2)$$

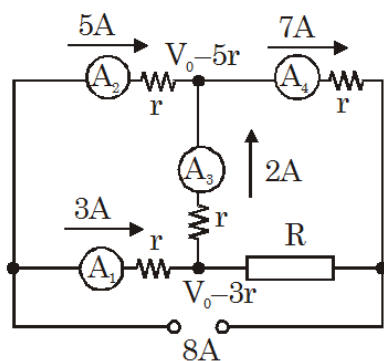
$$(2) - (1)$$

$$\Rightarrow IX - K \times 20 \quad \dots(3)$$

Divide (1) and (2)

$$\frac{R}{X} = \frac{1}{2} \Rightarrow x = 2R$$

3. **Ans (9.00)**



$$V_0 - 5r - 7r$$

$$= V_0 - 12r$$

$$V_0 - 3r - 1 \times R = V_0 - 12r$$

$$\Rightarrow R = 9r$$

4. **Ans (6.00)**

$$\frac{x-10}{10} + \frac{x+6}{20} + \frac{x}{10} + \frac{x-10}{5} = 0$$

$$\Rightarrow x = 6 \text{ V}$$

$$\therefore I_{20 \Omega} = \frac{6+6}{20} = 0.6 \text{ Amp}$$

5. **Ans (9.00)**

Energy released = $q.v$.

$$q = \text{area under curve} = \frac{1}{2} \times 150 \times 0.2 = \frac{30}{2} \text{ C}$$

$$v = E. d = 400 \times 10^3 \times 1500 = 60 \times 10^7$$

$$\text{energy released} = 60 \times 10^7 \times 15 = 900 \times 10^7 = 9 \times 10^9 \text{ J}$$

6. **Ans (2.00)**

$$V = \frac{kQ}{R} = \frac{\sigma \times 2\pi R^2}{4\pi\epsilon_0 R} = \frac{\sigma R}{2\epsilon_0}$$

8. **Ans (4.00)**

$$P = 2 \times \left(\frac{T}{R_1} + \frac{T}{R_2} \right) = 2 \times \left(\frac{T}{5R} + \frac{T}{R} \right) = \frac{12T}{5R}$$

9. **Ans (2.00)**

In steady state current in the branch containing the capacitor is zero & hence emf E is shared between

r & R_2 = in the ratio of their resistance voltage

$$\text{across } R_2 \text{ is } \frac{ER_2}{R_2 + r} = 2 \text{ Volts}$$

= Voltage across capacitor.

$$\therefore Q = CV_C = 2\mu\text{C}.$$

10. **Ans (3.00)**

$$V_C = \frac{4CV - CV}{3C} = V$$

$$\therefore U_f = \frac{1}{2} \times (C + 2C)V^2 = \frac{3}{2} CV^2$$

CHEMISTRY

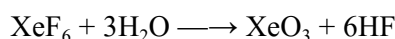
SEC-A

4. **Ans (C)**

The reaction is exothermic ($\Delta H < 0$)

$$\text{slope} = -\frac{\Delta H}{R}$$

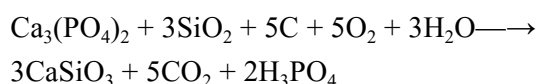
17. **Ans (C)**



CHEMISTRY

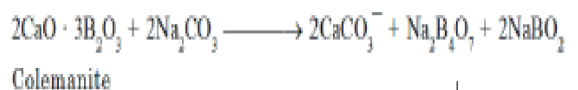
SEC-B

1. **Ans (0.63)**

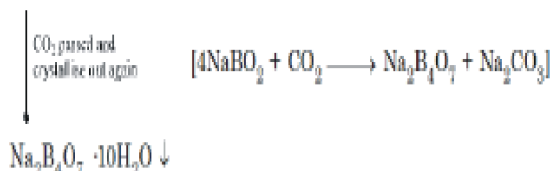
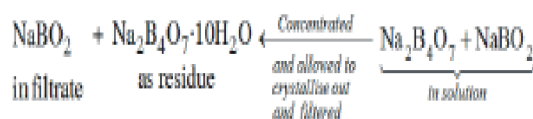


L.R.

4. **Ans (5.00)**



Filtered
- CaCO_3 (as residue)



$\therefore$ x is NaBO_2 & y is $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

Total number of Boron atoms = 1 + 4 = 5.00

5. **Ans (20.00)**

Let 'x' mole glucose be dissolved.

$$\left. \begin{aligned} 0.2 &= K_f \times \frac{x}{0.1} \\ 0.25 &= K_f \times \frac{x}{m} \end{aligned} \right\} m = 80 \text{ gm}$$

80 gm water is left hence, 20 gm ice will freeze out.

6. **Ans (19.00)**

$$112 - [\text{Rn}], 7s^2, 5f^{14}, 6d^{10}$$

Group number = 12, Period Number = 7

Answer is 12 + 7 = 19

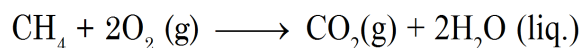
7. **Ans (4.00)**

$$P_{\text{N}_2} = 0.8 \times 5 = 4 \text{ atm}$$

$$P_{\text{N}_2} = K_H \times x$$

$$4 = 1 \times 10^5 \times \left(\frac{n_{\text{N}_2}}{10} \right) \Rightarrow n_{\text{N}_2} = 4 \times 10^{-4}$$

8. **Ans (100.00)**



80 ml

↓

80 ml

Total volume of CO_2 = 80 + 20 = 100 ml

9. **Ans (0.23)**

Moles of H_2SO_4 in initial sol = 0.1 mole

Mass of initial sol = 150 gm

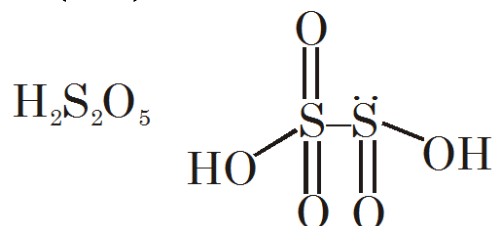
Mass of water = 400 gm

Total mass of sol = 550 gm

$$\text{Final volume} = \frac{550}{1.25} = 440 \text{ ml}$$

$$M = \frac{0.1}{0.44} = 0.227 \approx 0.23$$

10. **Ans (3.00)**



$$\text{S}-\text{S} \Rightarrow x = 1$$

$$\text{S}-\text{O}-\text{S} \Rightarrow y = 0$$

$$\text{O}-\text{H} \Rightarrow z = 2$$

$$\text{Ans.} (1 + 0 + 2 = 3)$$

MATHEMATICS

SEC-A

1. **Ans (A)**

$$(\alpha + 1)x + 2y + 5 = 0$$

$$-4x - \alpha y + 3 = 0$$

$$a_1a_2 + b_1b_2 > 0$$

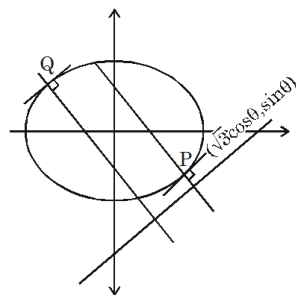
$$\Rightarrow -6\alpha > 4 \Rightarrow \alpha < -\frac{2}{3}$$

2. **Ans (A)**

A(1, 4), B(4, 1), C(-1, -4) and D(-1, 4)

Area of ΔABD = 3 sq. units

3. Ans (C)



$$\frac{x^2}{3} + \frac{y^2}{1} = 1$$

$$\frac{dy}{dx} = \frac{-x}{3y}$$

$$= \frac{-\sqrt{3} \cos \theta}{3 \sin \theta}$$

$$= \frac{-\cot \theta}{\sqrt{3}}$$

$$\frac{-\cot \theta}{\sqrt{3}} = 1 \Rightarrow \cot \theta = -\sqrt{3}$$

$$\Rightarrow \theta = -30^\circ \text{ or } 150^\circ$$

$$\text{For } Q, \theta = 150^\circ \Rightarrow Q \left(\frac{-3}{2}, \frac{1}{2} \right)$$

$$\text{Maximum distance} = \left| \frac{\frac{-3}{2} - \frac{1}{2} - 10}{\sqrt{2}} \right| = 6\sqrt{2}$$

4. Ans (C)

$$\left. \begin{array}{l} 1 + a + b = 0 \\ \text{and } c - 1 = 0 \end{array} \right\} c = 1 \text{ and } a + b = -1$$

$$\text{Now } \frac{dx_1}{dy_1} = 2cy + a = a + 2y$$

$$\left. \frac{dy_1}{dx_1} \right|_{(0,1)} = \frac{1}{a+2}$$

$$\text{and } \frac{dx}{dy} = c - 2y = 1 - 2y$$

$$\left. \frac{dy_2}{dx_2} \right|_{(0,1)} = \frac{1}{1-2} = -1$$

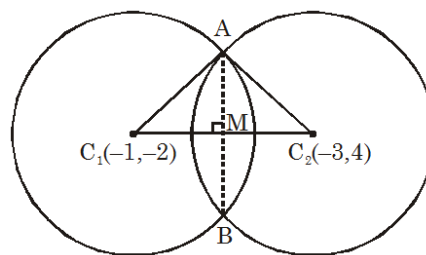
$$\left. \frac{dy_1}{dx_1} \right|_{(0,1)} \left(\frac{dy_2}{dx_2} \right) = -1$$

$$\Rightarrow \left(\frac{1}{a+2} \right) (-1) = -1$$

$$\Rightarrow a + 2 = 1 \Rightarrow \boxed{a = -1} \quad b = 0$$

$$2a + b - c = -2 + 0 + 1 = -1$$

5. Ans (A)



$$r_1 = 5, \quad r_2 = \sqrt{15}$$

$$C_1C_2 = \sqrt{4 + 36} = \sqrt{40}$$

$$\text{common chord } S_1 - S_2 = 0$$

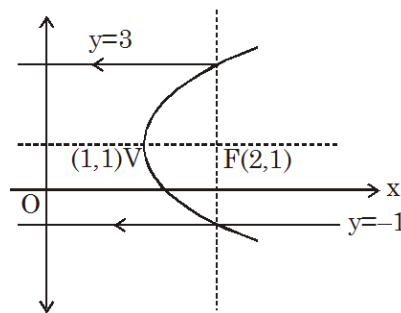
$$\Rightarrow 2x - 6y + 15 = 0$$

$$C_1M = p = \left| \frac{-2 + 12 + 15}{\sqrt{40}} \right| = \frac{25}{\sqrt{40}}$$

$$AB = 2\sqrt{25 - \frac{625}{40}} = 2\sqrt{\frac{375}{40}}$$

$$= 2\sqrt{\frac{75}{8}} = 5\sqrt{\frac{3}{2}}$$

6. Ans (C)



$$(y - 1)^2 = 4x - 4$$

$$(y - 1)^2 = 4(x - 1)$$

$$y - 1 = \pm 2$$

$$\text{Equation of reflected rays}$$

$$y = -1 \text{ and } y = 3$$

7. Ans (A)

$$\frac{b}{a} = \frac{3}{2}, \text{ Now}$$

$$\frac{2ae}{\frac{2a}{e}} = e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{9}{4} = \frac{13}{4}$$

8. Ans (A)

$$\vec{a} \cdot \vec{b} = 0 \text{ and } \vec{a} \cdot \vec{c} = 0$$

$$|\vec{a} + \vec{b} + \vec{c}|^2 = 1 + 1 + 1 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 1$$

$$\Rightarrow \vec{c} \cdot \vec{a} = -1 \Rightarrow \vec{c} \wedge \vec{a} = \pi$$

9. **Ans (A)**

$$P(5 \cos \theta, 2 \sin \theta), Q(5 \sin \theta, -2 \cos \theta)$$

$$PQ = \sqrt{(5 \cos \theta - 5 \sin \theta)^2 + (2 \sin \theta + 2 \cos \theta)^2}$$

$$PQ = \sqrt{29 - 21 \sin 2\theta}$$

$$\text{Minimum value of } PQ = \sqrt{8} = 2\sqrt{2}$$

10. **Ans (B)**

$$D = \begin{vmatrix} a+1 & -1 & -1 \\ 1 & -(a+1) & -1 \\ 1 & -1 & -(a+1) \end{vmatrix} = a^2(a+3)$$

$$\Rightarrow a^2(a+3) = 0 \Rightarrow a = 0, -3$$

for $a = 0$ there are infinite solutions and for $a = -3$, there is no solution.

11. **Ans (A)**

$$\text{Let equation of plane } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

passes through $(2, 2, 2)$

$$\text{i.e. } \frac{2}{a} + \frac{2}{b} + \frac{2}{c} = 1 \Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{2}$$

$$\alpha = \frac{a}{4} \Rightarrow a = 4\alpha$$

$$\text{Similarly } b = 4\beta \text{ and } c = 4\gamma$$

$$\Rightarrow \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 2$$

For three positive numbers α, β and γ

A.M. $\geq$ H.M.

$$\frac{\alpha + \beta + \gamma}{3} \geq \frac{3}{\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}} \Rightarrow \alpha + \beta + \gamma \geq \frac{9}{2}$$

12. **Ans (C)**

$\vec{p} - \vec{q}$, $2\vec{q} + \vec{p}$ and $3\vec{p} - \vec{q}$ are coplanar

$$\text{L.H.S.} = 0 \Rightarrow \text{R.H.S.} = 0 \Rightarrow \vec{p} = -\vec{q}$$

$$\vec{p} \wedge \vec{q} = \pi$$

13. **Ans (B)**

$$(\vec{p} \times \vec{q}) \cdot (\vec{p} \times \vec{q}) = \frac{1}{2}$$

$$\Rightarrow \sin^2 \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{4}$$

14. **Ans (B)**

$$\text{Let } \vec{x} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & c \\ 1 & -2 & 1 \end{vmatrix} = (b+2c)\hat{i} - \hat{j}(a-c) + \hat{k}(-2a-b)$$

$$= -\hat{i} + \hat{k}$$

$$b+2c = -1, a-c = 0 \text{ and } 2a+b = -1$$

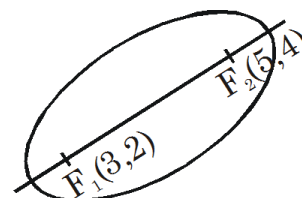
$$\text{and } a^2 + b^2 + c^2 = 1$$

$$\Rightarrow a = 0, b = -1, c = 0$$

$$\text{or } a = -\frac{2}{3}, b = \frac{1}{3}, c = -\frac{2}{3}$$

$$\Rightarrow \vec{x} = -\hat{j} \text{ or } \vec{x} = -\frac{1}{3}(2\hat{i} - \hat{j} + 2\hat{k})$$

15. **Ans (A)**



$$2ae = 2\sqrt{2}$$

$$\Rightarrow ae = \sqrt{2}$$

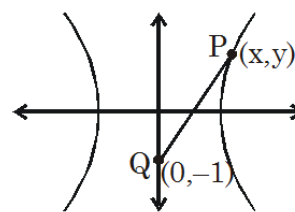
$$2a = 6 \Rightarrow a = 3$$

$$\Rightarrow c = \frac{\sqrt{2}}{3}$$

Distance between directrices

$$= 2 \left(\frac{3}{\frac{\sqrt{2}}{3}} \right) = 9\sqrt{2}$$

16. **Ans (B)**



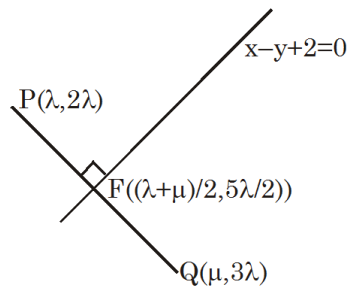
$$PQ = \sqrt{x^2 + (y+1)^2}$$

$$= \sqrt{y^2 + 4 + y^2 + 2y + 1}$$

$$= \sqrt{2\left(y + \frac{1}{2}\right)^2 + \frac{9}{2}}$$

$$PQ \text{ is minimum if } y = -\frac{1}{2}$$

17. Ans (B)



$$\frac{\lambda + \mu}{2} - \frac{5\lambda}{2} + 2 = 0$$

$$\mu - 4\lambda + 4 = 0 \quad \dots(1)$$

$$\text{and } m_{PQ} = \frac{\lambda}{\mu - \lambda} = -1 \quad \dots(2)$$

$$\Rightarrow \boxed{\mu = 0} \text{ and } \boxed{\lambda = 1}$$

18. Ans (C)

$$1 - 4m - m^2 = 0 \Rightarrow m^2 + 4m - 1 = 0$$

$$m_1 + m_2 = -4 \text{ and } m_1 m_2 = -1$$

$$\tan(\theta_1 + \theta_2) = \frac{-4}{1 + 1} = -2$$

$$\sec^2(\theta_1 + \theta_2) = 5$$

$$\text{and } \left| \frac{1}{\tan \theta_1} + \frac{1}{\tan \theta_2} \right| = \left| \frac{\tan \theta_1 + \tan \theta_2}{\tan \theta_1 + \tan \theta_2} \right| = 4$$

19. Ans (C)

$$(x)^3 + (2y)^3 + (-4)^3 = 3(x)(2y)(-4)$$

$$\Rightarrow x = 2y = -4 \Rightarrow x = -4, y = -2$$

i.e. a point $(-4, -2)$

or $x + 2y - 4 = 0$ i.e. a line

20. Ans (A)

$$\left[\begin{matrix} \vec{AB} & \vec{AC} & \vec{AD} \end{matrix} \right] = 0$$

$$\Rightarrow \begin{vmatrix} -1 & -1 & 0 \\ -\lambda - 2 & 0 & 0 \\ 1 & -2 & -2 \end{vmatrix} = 0$$

$$\Rightarrow \lambda = -2$$

MATHEMATICS

SEC-B

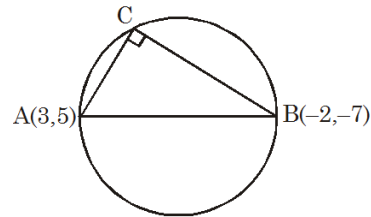
1. Ans (1.00)

R will be the mid point of PQ

$$\text{i.e. } R\left(4, \frac{3}{2}\right) \Rightarrow \alpha = 4, \beta = \frac{3}{2}$$

$$\alpha - 2\beta = 4 - 3 = 1$$

2. Ans (4.00)



$$AB = \sqrt{25 + 144} = 13$$

Maximum area of

$$\Delta ABC = \frac{1}{2}(13)\left(\frac{13}{2}\right) = \frac{169}{4}$$

So 4 such points C are possible.

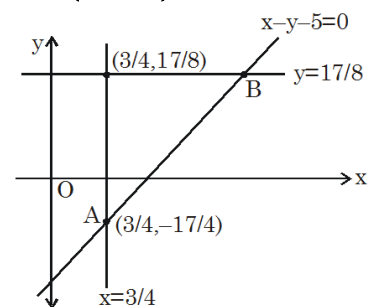
3. Ans (1.00)

$$3\alpha x + 2x + 18\alpha + 9 = 0$$

$$\Rightarrow x = \frac{-18\alpha - 9}{3\alpha + 2} = -6 + \frac{3}{3\alpha + 2}$$

Only possible integral value of α is -1 .

4. Ans (59.00)



Locus of centre will be angle bisectors of given

lines i.e. $x = \frac{3}{4}$ and $y = \frac{17}{8}$

$$\text{Required area} = \frac{1}{2} \left(\frac{51}{8} \right)^2 \Rightarrow P = 51, Q = 8$$

$$P + Q = 59$$

5. Ans (29.00)

$$\vec{r}_1 = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\vec{r}_2 = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$|\vec{r}_1 \times \vec{r}_2| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & c \\ 2 & 3 & 4 \end{vmatrix}$$

$$= |\hat{i}(4b - 3c) - \hat{j}(4a - 2c) + \hat{k}(3a - 2b)|$$

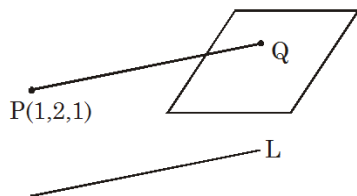
$$= \left(\sqrt{a^2 + b^2 + c^2} \right) (\sqrt{29}) \sin \theta$$

$\Rightarrow$

$$(4b - 3c)^2 + (4a - 2c)^2 + (3a - 2b)^2 = 29 \sin^2 \theta$$

$$\text{Max. value} = 29$$

6. Ans (6.06)



Equation of line PQ

$$\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-1}{1} = \lambda \text{ (say)}$$

$$Q(\lambda+1, \lambda+2, \lambda+1).$$

Now Q lies on the plane

$$\text{i.e. } 2(\lambda+1) + (\lambda+2) - (\lambda+1) = 10$$

$$\Rightarrow 2\lambda = 7 \Rightarrow \lambda = \frac{7}{2} \Rightarrow Q\left(\frac{9}{2}, \frac{11}{2}, \frac{9}{2}\right)$$

$$PQ = \sqrt{\frac{49}{4} + \frac{49}{4} + \frac{49}{4}} = \frac{7}{2}\sqrt{3}$$

7. Ans (5.05, 5.06)

$$\begin{vmatrix} 2 & 2 & \lambda - \frac{7}{2} \\ 3 & -1 & 2 \\ -2 & 3\lambda & \frac{1}{2} \end{vmatrix} = 0$$

$$\Rightarrow 2\left(-\frac{1}{2} - 6\lambda\right) - 2\left(\frac{3}{2} + 4\right) + \left(\lambda - \frac{7}{2}\right)(9\lambda - 2) = 0$$

$$\Rightarrow 18\lambda^2 - 91\lambda - 10 = 0$$

$$\Rightarrow \lambda_1 + \lambda_2 = \frac{91}{18} = \frac{182}{36}$$

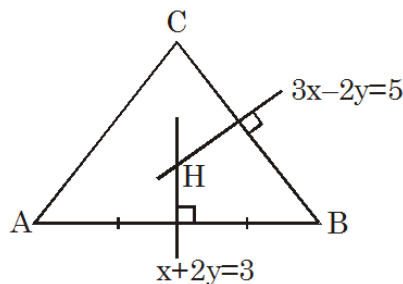
8. Ans (1.00)

Let foot of $\perp$ be P(x, y, z) then $\vec{AP} \cdot \vec{PB} = 0$
 $\Rightarrow (x-0)(x-1) + (y-1)(y+2) + (z-2)(z-3) = 0$

$$x^2 + y^2 + z^2 - x + y - 5z + 4 = 0$$

$$\alpha + \beta + \gamma + \delta = -1 + 1 - 5 + 4 = -1$$

9. Ans (1.66, 1.67)



Lines $3x - 2y = 5$ and $x + 2y = 3$ are perpendicular bisectors of the triangle ABC

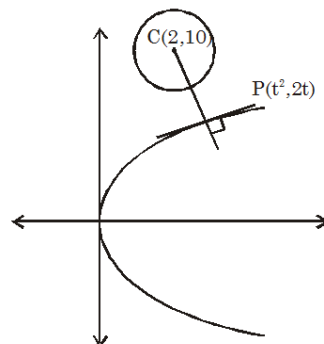
Hence circumcentre H $\left(2, \frac{1}{2}\right)$ and point P is centroid.

$$\vec{OP} = \frac{2}{3}\vec{OH} \Rightarrow P\left(\frac{4}{3}, \frac{1}{3}\right)$$

$$\Rightarrow a + b = \frac{5}{3}$$

10. Ans (0.25)

$$\text{Radius} = \sqrt{4 + 100} - 10 = 1$$



$$\text{Slope of CP} = \frac{-1}{\frac{dy}{dx} \Big|_{\text{at } t}} = -t = \frac{2t-10}{t^2-2}$$

$$\Rightarrow -t^3 + 2t = 2t - 10 \Rightarrow t = (10)^{1/3}$$

$$P\left((10)^{2/3}, 2(10)^{1/3}\right)$$

$$\Rightarrow \alpha = (10)^{2/3}, \beta = 2(10)^{1/3}$$

$$\alpha\beta = 2(10) = 20$$